

Internal Dependencies

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [Neo4j Python Driver](#)

Artifacts

List the artifacts this notebook is based on. Different sorting variations help finding artifacts by their features and support larger code bases where the list of all artifacts gets too long.

Only the top 30 entries are shown. The whole table can be found in the following CSV report:

List_all_Java_artifacts

Table 1a - Top 30 artifacts with the highest package count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-5.0.1.jar	57	570	7	2
1	axon-common-5.0.1.jar	13	150	10	0
2	axon-eventsourcing-5.0.1.jar	7	100	3	4
3	axon-modelling-5.0.1.jar	7	93	2	3
4	axon-spring-boot-autoconfigure-5.0.1.jar	7	70	0	7
5	axon-server-connector-5.0.1.jar	5	72	1	4
6	axon-test-5.0.1.jar	5	73	1	3
7	axon-update-5.0.1.jar	5	23	1	1
8	axon-conversion-5.0.1.jar	4	30	4	1
9	axon-metrics-micrometer-5.0.1.jar	2	13	0	2
10	axon-tracing-opentelemetry-5.0.1.jar	1	5	0	2

Table 1b - Top 30 artifacts with the highest type count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-5.0.1.jar	57	570	7	2
1	axon-common-5.0.1.jar	13	150	10	0
2	axon-eventsourcing-5.0.1.jar	7	100	3	4
3	axon-modelling-5.0.1.jar	7	93	2	3
4	axon-test-5.0.1.jar	5	73	1	3
5	axon-server-connector-5.0.1.jar	5	72	1	4
6	axon-spring-boot-autoconfigure-5.0.1.jar	7	70	0	7
7	axon-conversion-5.0.1.jar	4	30	4	1
8	axon-update-5.0.1.jar	5	23	1	1
9	axon-metrics-micrometer-5.0.1.jar	2	13	0	2
10	axon-tracing-opentelemetry-5.0.1.jar	1	5	0	2

Table 1c - Top 30 artifacts with the highest number of incoming dependencies

The following table lists the top 30 artifacts that are used the most by other artifacts (highest count of incoming dependencies, highest in-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-common-5.0.1.jar	13	150	10	0
1	axon-messaging-5.0.1.jar	57	570	7	2
2	axon-conversion-5.0.1.jar	4	30	4	1
3	axon-eventsourcing-5.0.1.jar	7	100	3	4
4	axon-modelling-5.0.1.jar	7	93	2	3
5	axon-server-connector-5.0.1.jar	5	72	1	4
6	axon-test-5.0.1.jar	5	73	1	3
7	axon-update-5.0.1.jar	5	23	1	1
8	axon-metrics-micrometer-5.0.1.jar	2	13	0	2
9	axon-spring-boot-autoconfigure-5.0.1.jar	7	70	0	7
10	axon-tracing-opentelemetry-5.0.1.jar	1	5	0	2

Table 1d - Top 30 artifacts with the highest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the highest number of other artifacts (highest count of outgoing dependencies, highest out-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-spring-boot-autoconfigure-5.0.1.jar	7	70	0	7
1	axon-eventsourcing-5.0.1.jar	7	100	3	4
2	axon-server-connector-5.0.1.jar	5	72	1	4
3	axon-modelling-5.0.1.jar	7	93	2	3
4	axon-test-5.0.1.jar	5	73	1	3
5	axon-messaging-5.0.1.jar	57	570	7	2
6	axon-metrics-micrometer-5.0.1.jar	2	13	0	2
7	axon-tracing-opentelemetry-5.0.1.jar	1	5	0	2
8	axon-conversion-5.0.1.jar	4	30	4	1
9	axon-update-5.0.1.jar	5	23	1	1
10	axon-common-5.0.1.jar	13	150	10	0

Table 1e - Top 30 artifacts with the lowest package count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-tracing-opentelemetry-5.0.1.jar	1	5	0	2
1	axon-metrics-micrometer-5.0.1.jar	2	13	0	2
2	axon-conversion-5.0.1.jar	4	30	4	1
3	axon-server-connector-5.0.1.jar	5	72	1	4
4	axon-test-5.0.1.jar	5	73	1	3
5	axon-update-5.0.1.jar	5	23	1	1
6	axon-eventsourcing-5.0.1.jar	7	100	3	4
7	axon-modelling-5.0.1.jar	7	93	2	3
8	axon-spring-boot-autoconfigure-5.0.1.jar	7	70	0	7
9	axon-common-5.0.1.jar	13	150	10	0
10	axon-messaging-5.0.1.jar	57	570	7	2

Table 1f - Top 30 artifacts with the lowest type count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-tracing-opentelemetry-5.0.1.jar	1	5	0	2
1	axon-metrics-micrometer-5.0.1.jar	2	13	0	2
2	axon-update-5.0.1.jar	5	23	1	1
3	axon-conversion-5.0.1.jar	4	30	4	1
4	axon-spring-boot-autoconfigure-5.0.1.jar	7	70	0	7
5	axon-server-connector-5.0.1.jar	5	72	1	4
6	axon-test-5.0.1.jar	5	73	1	3
7	axon-modelling-5.0.1.jar	7	93	2	3
8	axon-eventsourcing-5.0.1.jar	7	100	3	4
9	axon-common-5.0.1.jar	13	150	10	0
10	axon-messaging-5.0.1.jar	57	570	7	2

Table 1g - Top 30 artifacts with the lowest number of incoming dependencies

The following table lists the top 30 artifacts that are used the least by other artifacts (lowest count of incoming dependencies, lowest in-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-metrics-micrometer-5.0.1.jar	2	13	0	2
1	axon-spring-boot-autoconfigure-5.0.1.jar	7	70	0	7
2	axon-tracing-opentelemetry-5.0.1.jar	1	5	0	2
3	axon-server-connector-5.0.1.jar	5	72	1	4
4	axon-test-5.0.1.jar	5	73	1	3
5	axon-update-5.0.1.jar	5	23	1	1
6	axon-modelling-5.0.1.jar	7	93	2	3
7	axon-eventsourcing-5.0.1.jar	7	100	3	4
8	axon-conversion-5.0.1.jar	4	30	4	1
9	axon-messaging-5.0.1.jar	57	570	7	2
10	axon-common-5.0.1.jar	13	150	10	0

Table 1h - Top 30 artifacts with the lowest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the lowest number of other artifacts (lowest count of outgoing dependencies, lowest out-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-common-5.0.1.jar	13	150	10	0
1	axon-conversion-5.0.1.jar	4	30	4	1
2	axon-update-5.0.1.jar	5	23	1	1
3	axon-messaging-5.0.1.jar	57	570	7	2
4	axon-metrics-micrometer-5.0.1.jar	2	13	0	2
5	axon-tracing-opentelemetry-5.0.1.jar	1	5	0	2
6	axon-modelling-5.0.1.jar	7	93	2	3
7	axon-test-5.0.1.jar	5	73	1	3
8	axon-eventsourcing-5.0.1.jar	7	100	3	4
9	axon-server-connector-5.0.1.jar	5	72	1	4
10	axon-spring-boot-autoconfigure-5.0.1.jar	7	70	0	7

Cyclic Dependencies

Cyclic dependencies occur when one package uses a class of another package and vice versa. These dependencies can lead to problems when one of these packages needs to be changed.

Table 2a - Cyclic Dependencies Overview

Show the top 40 cyclic dependencies sorted by the most promising to resolve first. This is done by calculating the number of forward dependencies (first cycle participant to second cycle participant) in relation to backward dependencies (second cycle participant back to first cycle participant). The higher this rate (approaching 1), the easier it should be to resolve the cycle by focussing on the few backward dependencies.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic_Dependencies](#)

Columns:

- *artifactName* identifies the artifact of the first participant of the cycle
- *packageName* identifies the package of the first participant of the cycle
- *dependentArtifactName* identifies the artifact of the second participant of the cycle
- *dependentPackageName* identifies the package of the second participant of the cycle
- *forwardToBackwardBalance* is between 0 and 1. High for many forward and few backward dependencies.
- *numberForward* contains the number of dependencies from the first participant of the cycle to the second one
- *numberBackward* contains the number of dependencies from the second participant of the cycle back to the first one
- *someForwardDependencies* lists some forward dependencies in the text format "type1 -> type2"
- *backwardDependencies* lists the backward dependencies in the format "type1 <- type2" that are recommended to get resolved

	artifactName	packageName	dependentArtifactName	dependentPackageName	forwardToBackwardBalance	numberForward	numberBack
0	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core	0.959184	48	
1	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	axon-messaging-5.0.1	org.axonframework.messaging.core	0.942857	34	
2	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	0.933333	29	
3	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	0.894737	18	
4	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	0.875000	15	
5	axon-modelling-5.0.1	org.axonframework.modelling.annotation	axon-modelling-5.0.1	org.axonframework.modelling	0.846154	12	
6	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	0.666667	15	
7	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	0.666667	15	
8	axon-common-5.0.1	org.axonframework.common.configuration	axon-common-5.0.1	org.axonframework.common.infra	0.529412	13	
9	axon-messaging-5.0.1	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	0.500000	15	
10	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.transaction	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	0.500000	6	
11	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	0.428571	5	
12	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.command	0.333333	4	
13	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	0.333333	2	
14	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.sequencing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	0.333333	8	
15	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	0.200000	6	
16	axon-modelling-5.0.1	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.1	org.axonframework.modelling.annotation	0.200000	3	
17	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	axon-messaging-5.0.1	org.axonframework.messaging.core	0.157895	11	
18	axon-modelling-5.0.1	org.axonframework.modelling.entity.child	axon-modelling-5.0.1	org.axonframework.modelling.entity	0.142857	8	

	artifactName	packageName	dependentArtifactName	dependentPackageName	forwardToBackwardBalance	numberForward	numberBack
19	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.util	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	0.000000	2	
20	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.annotation.reflection	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.annotation	0.000000	1	

Table 2b - Cyclic Dependencies Break Down

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic_Dependencies_Breakdown

Columns in addition to Table 2a:

- dependency shows the cycle dependency in the text format "type1 -> type2" (forward) or "type2<-type1" (backward)

[illegible]

Table 2c - Cyclic Dependencies Break Down - Backward Dependencies Only

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first. This table only contains the backward dependencies from the second participant of the cycle back to the first one that are the most promising to resolve.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic_Dependencies_Breakdown_BackwardOnly

	artifactName	packageName	dependentArtifactName	dependentPackageName	dependency forName
0	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core	HandlerAttributes<-SimpleHandlerAttributes
1	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	axon-messaging-5.0.1	org.axonframework.messaging.core	EventMessage<-SubscribableEventSource
2	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	EventHandler<-HandlerTypeResolver
3	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	CommandHandler<-HandlerTypeResolver
4	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	QueryHandler<-HandlerTypeResolver
5	axon-modelling-5.0.1	org.axonframework.modelling.annotation	axon-modelling-5.0.1	org.axonframework.modelling	TargetEntityIdMemberMismatchException<-PropertyBasedEntityIdResolver
6	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorModule<-EventProcessorModule
7	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorConfiguration<-EventProcessorModule
8	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorsConfigurer<-EventProcessingConfigurer
9	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorsConfigurer<-EventProcessingConfigurer
10	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorModule<-EventProcessorModule
11	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorConfiguration<-EventProcessorModule
12	axon-common-5.0.1	org.axonframework.common.configuration	axon-common-5.0.1	org.axonframework.common.infra	Component<-FilesystemStyleComponentDescriptor
13	axon-common-5.0.1	org.axonframework.common.configuration	axon-common-5.0.1	org.axonframework.common.infra	Component\$Identifier<-FilesystemStyleComponentDescriptor
14	axon-common-5.0.1	org.axonframework.common.configuration	axon-common-5.0.1	org.axonframework.common.infra	Component\$Identifier<-JacksonComponentDescriptor
15	axon-common-5.0.1	org.axonframework.common.configuration	axon-common-5.0.1	org.axonframework.common.infra	Component<-JacksonComponentDescriptor
16	axon-messaging-5.0.1	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	MessageHandlerInterceptorMemberChain<-ChainedMessageHandlerInterceptorMember
17	axon-messaging-5.0.1	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	MessageInterceptingMember<-AnnotatedHandlerInspector
18	axon-messaging-5.0.1	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	MessageHandlerInterceptorMemberChain<-AnnotatedHandlerInspector
19	axon-messaging-5.0.1	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	NoMoreInterceptors<-AnnotatedHandlerInspector
20	axon-messaging-5.0.1	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.1	org.axonframework.messaging.core.annotation	NoMoreInterceptors<-ChainedMessageHandlerInterceptorMember
21	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.transaction	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	TransactionManager<-TransactionalUnitOfWorkFactory
22	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.transaction	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	Transaction<-TransactionalUnitOfWorkFactory
23	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	EventProcessorControlService<-AxonServerConfigurationEnhancer
24	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	AxonServerEventStorageEngineFactory<-AxonServerConfigurationEnhancer
25	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.command	MetadataConverter<-CommandConverter
26	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.command	AxonServerConfiguration<-AxonServerCommandBusConnector
27	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.configuration	MessagingConfigurer<-EventProcessingConfigurer
28	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.sequencing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	SequentialPolicy<-SimpleEventHandlingComponent
29	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.sequencing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	HierarchicalSequencingPolicy<-SimpleEventHandlingComponent
30	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.sequencing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	SequentialPerAggregatePolicy<-SimpleEventHandlingComponent
31	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.sequencing	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling	SequencingPolicy<-SimpleEventHandlingComponent
32	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	AxonServerNonTransientRemoteQueryHandlingException<-ErrorCode
33	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	AxonServerRemoteQueryHandlingException<-ErrorCode
34	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	AxonServerQueryDispatchException<-ErrorCode
35	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.1	org.axonframework.axonserver.connector	AxonServerQueryBusConnector<-AxonServerConfigurationEnhancer
36	axon-modelling-5.0.1	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.1	org.axonframework.modelling.annotation	AnnotatedEntityMetamodel<-EntityIdResolverDefinition
37	axon-modelling-5.0.1	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.1	org.axonframework.modelling.annotation	AnnotatedEntityMetamodel<-AnnotationBasedEntityIdResolverDefinition
38	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	axon-messaging-5.0.1	org.axonframework.messaging.core	ProcessingContext<-MessageHandlerInterceptor
39	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork	axon-messaging-5.0.1	org.axonframework.messaging.core	ProcessingContext<-Message

Interface Segregation Candidates

Well known from [Design Principles and Design Patterns by Robert C. Martin](#), the *Interface Segregation Principle* suggests that software components should have narrow, focused interfaces rather than large, general-purpose ones. The goal is to minimize the dependencies between components and increase modularity, flexibility, and maintainability.

Smaller, focused and purpose-driven interfaces

- make it easier to modify individual components without affecting the rest of the system.
- make it clearer which client is affected by which change.
- don't force their clients to depend on methods they don't need.
- reduce the scope of changes since a change to one component doesn't affect others.
- lead to a more loosely coupled architecture that is easier to understand and maintain.

Reference: [Analyze java package metrics in a graph database](#)

How to apply the results

If just one method of a type is used, especially in many places, then the result of this method can be used to call e.g. a method or construct an object instead of using the whole object and then just calling that single method.

If there are a couple of methods that are used for a distinct purpose, those could be factored out into a separate interface. The original type can extended/implement the new interface so that there are no breaking changes. Then all the callers, that use only this group of methods, can be changed to the new interface.

Table 4 - Top 40 most used combinations of methods

The following table shows the top 40 most used combinations of methods of larger types that might benefit from applying the *Interface Segregation Principle*. The whole table can be found in the CSV report `Candidates_for_Interface_Segregation`.

	fullDependentTypeName	declaredMethods	calledMethodNames	calledMethods	callerTypes
0	org.axonframework.messaging.core.unitofwork.ProcessingContext	32	[computeResourceIfAbsent]	1	7
1	org.axonframework.messaging.core.unitofwork.UnitOfWork	24	[executeWithResult]	1	6
2	org.axonframework.messaging.commandhandling.CommandBus	6	[dispatch]	1	6
3	org.axonframework.common.configuration.ComponentDefinition\$ComponentCreator	17	[createComponent]	1	5
4	org.axonframework.messaging.core.unitofwork.ProcessingContext	32	[withResource]	1	4
5	org.axonframework.messaging.eventhandling.EventMessage	19	[timestamp]	1	4
6	org.axonframework.messaging.eventhandling.EventMessage	19	[identifier, timestamp]	2	4
7	org.axonframework.messaging.core.MessageStream\$Entry	7	[message]	1	4
8	org.axonframework.messaging.core.MessageStream\$Empty	44	[cast]	1	3
9	org.axonframework.messaging.core.DelayedMessageStream	42	[create]	1	3
10	org.axonframework.messaging.core.unitofwork.UnitOfWork	24	[execute]	1	3
11	org.axonframework.messaging.eventhandling.EventMessage	19	[identifier]	1	3
12	org.axonframework.messaging.core.annotation.WrappedMessageHandlingMember	13	[handleSync]	1	3
13	org.axonframework.messaging.core.unitofwork.ProcessingContext	32	[putResource]	1	2
14	org.axonframework.messaging.core.unitofwork.UnitOfWork	25	[executeWithResult]	1	2
15	org.axonframework.common.configuration.AbstractComponent	24	[describeTo]	1	2
16	org.axonframework.test.fixture.AxonTestThenMessage	22	[exceptionSatisfies, exception]	3	2
17	org.axonframework.messaging.commandhandling.CommandMessage	19	[andMetadata]	1	2
18	org.axonframework.messaging.eventhandling.EventMessage	19	[withConvertedPayload]	1	2
19	org.axonframework.messaging.eventhandling.EventMessage	19	[andMetadata]	1	2
20	org.axonframework.messaging.eventstreaming.EventCriterion	12	[tags]	1	2
21	org.axonframework.messaging.eventstreaming.OrEventCriteria	12	[or]	1	2
22	org.axonframework.modelling.entity.ConcreteEntityMetamodel	11	[forEntityTypeClass]	1	2
23	org.axonframework.modelling.entity.PolymorphicEntityMetamodel	11	[forSuperType]	1	2
24	org.axonframework.modelling.entity.annotation.AnnotatedEntityMetamodel	11	[entityType]	1	2
25	org.axonframework.eventsourcing.eventstore.EventStore	9	[transaction]	1	2
26	org.axonframework.messaging.eventhandling.EventHandlingComponent	8	[supportedEvents, sequenceIdentifierFor, supports]	3	2
27	org.axonframework.messaging.eventhandling.processing.streaming.token.WrappedToken	8	[unwrapLowerBound]	1	2
28	org.axonframework.messaging.core.MessageStream\$Entry	7	[map]	1	2
29	org.axonframework.messaging.core.unitofwork.UnitOfWork\$UnitOfWorkProcessingContext	45	[whenComplete, isError, isStarted, on, isCommitted, commit, onError, isCompleted]	8	1
30	org.axonframework.messaging.core.EmptyMessageStream	44	[instance]	1	1
31	org.axonframework.eventsourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream	43	[isCompleted, hasNextAvailable]	2	1
32	org.axonframework.eventsourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream	42	[callback]	1	1
33	org.axonframework.messaging.core.DelayedMessageStream	42	[createSingle]	1	1
34	org.axonframework.messaging.core.MessageStream\$Single	42	[cast]	1	1
35	org.axonframework.messaging.core.MessageStream\$Single	42	[asCompletableFuture]	1	1
36	org.axonframework.messaging.core.MessageStream\$Single	42	[first]	1	1
37	org.axonframework.messaging.core.unitofwork.ProcessingContext	37	[updateResource, removeResource, putResourceIfAbsent, putResource, computeResourceIfAbsent]	6	1
38	org.axonframework.messaging.core.unitofwork.ProcessingContext	36	[updateResource, computeResourceIfAbsent]	2	1
39	org.axonframework.messaging.core.unitofwork.ProcessingContext	35	[computeResourceIfAbsent]	1	1

Package Usage

Table 5 - Types that are used by multiple packages

This table shows the top 40 packages that are used by the highest number of different packages. The whole table can be found in the CSV report `List_types_that_are_used_by_many_different_packages` .

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
0	org.axonframework.messaging.core.unitofwork.ProcessingContext	ProcessingContext	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...	57
1	org.axonframework.common.annotation.Internal	Internal	[Type, File, Java, ByteCode, Annotation, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent...	47
2	org.axonframework.messaging.core.Message	Message	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...	46
3	org.axonframework.common.infra.ComponentDescriptor	ComponentDescriptor	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...	39
4	org.axonframework.messaging.eventhandling.EventMessage	EventMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedCompen...	35
5	org.axonframework.messaging.core.MessageStream	MessageStream	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWea...	35
6	org.axonframework.messaging.core.MessageType	MessageType	[Type, File, Java, ByteCode, Record, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, M...	29
7	org.axonframework.messaging.core.QualifiedName	QualifiedName	[Type, File, Java, ByteCode, Record, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4T...	29
8	org.axonframework.common.configuration.Configuration	Configuration	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation24, Mark4TypeLouvainCommunity1...	28
9	org.axonframework.common.FutureUtils	FutureUtils	[Type, File, Java, Class, ByteCode, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation14, Mark4TypeLouvainCommunity2, Mark...	26
10	org.axonframework.common.infra.DescribableComponent	DescribableComponent	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...	24
11	org.axonframework.messaging.core.MessageStream\$Empty	MessageStream\$Empty	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation47, Mark4TypeLouvainCommunity18, Mark4TypeLeidenCommunity4, Mark4Type...	24
12	org.axonframework.messaging.core.Context\$ResourceKey	Context\$ResourceKey	[Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyCon...	23
13	org.axonframework.common.BuilderUtils	BuilderUtils	[Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity11, Mark4TypeKCoreDecomposition8, Mark4TypeMaximumKCut43, Mark4TypeHDBSCAN72, Mark4TopAnomalyHub]	21
14	org.axonframework.common.configuration.ComponentRegistry	ComponentRegistry	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation33, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity7, Mark4TypeKCoreDecomposition8, Mark4TypeMaximumKCut16, Mark4TypeHDBSCAN61]	21
15	org.axonframework.common.Assert	Assert	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCostEffectivenessLazyForward, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity5, Mark4Typ...	20
16	org.axonframework.messaging.commandhandling.CommandMessage	CommandMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity15, Mark4TypeLeidenCommunity13...	20
17	org.axonframework.messaging.core.MessageStream\$Entry	MessageStream\$Entry	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub,	19

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
			Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation47, Mark4TypeLouvainCommunity18, Mark4...	
18	org.axonframework.messaging.core.annotation.ParameterResolverFactory	ParameterResolverFactory	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation66, Mark4TypeLouvainCommunity10, Mark4TypeLeidenCommunity11, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut95, Mark4TypeHDBSCAN121]	19
19	org.axonframework.messaging.core.MessageStream\$Single	MessageStream\$Single	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation47, Mar...	18
20	org.axonframework.common.configuration.ConfigurationEnhancer	ConfigurationEnhancer	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation28, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN-1]	17
21	org.axonframework.messaging.core.Metadata	Metadata	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut0, Mar...	16
22	org.axonframework.conversion.Converter	Converter	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecompo...	15
23	org.axonframework.messaging.core.conversion.MessageConverter	MessageConverter	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity10, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut10, Mark4TypeLocalClusteringCoefficient0.1225296442687747, Mark4TypeHDBSCAN-1]	15
24	org.axonframework.messaging.eventhandling.processing.streaming.token.TrackingToken	TrackingToken	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation48, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, ...	15
25	org.axonframework.common.AxonConfigurationException	AxonConfigurationException	[Type, File, Java, Class, ByteCode, Throwable, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation10, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut73, Mark4TypeHDBSCAN-1]	14
26	org.axonframework.messaging.core.MessageTypeResolver	MessageTypeResolver	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut79, Mark4TypeHDBSCAN-1]	14
27	org.axonframework.common.ObjectUtils	ObjectUtils	[Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN111]	14
28	org.axonframework.messaging.commandhandling.CommandBus	CommandBus	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity15, Mark4TypeLeidenCommunity13, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut51, Mark4TypeHDBSCAN-1]	13
29	org.axonframework.messaging.queryhandling.QueryMessage	QueryMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation44, Mark4TypeLouvainCommunity13, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition9, ...	13
30	org.axonframework.messaging.commandhandling.CommandResultMessage	CommandResultMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity13, Mark4TypeCoreDecomposition9, ...	12
31	org.axonframework.common.configuration.ComponentDefinition	ComponentDefinition	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation37, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut52, Mark4TypeLocalClusteringCoefficient0.2028985507246377, Mark4T...	12
32	org.axonframework.common.configuration.ComponentDefinition\$IncompleteComponentDefinition	ComponentDefinition\$IncompleteComponentDefinition	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation37, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut40, Mark4TypeHDBSCAN139]	12
33	org.axonframework.messaging.core.Context	Context	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation47, Mark4TypeLouvainCommunity18, Mark4TypeLeiden...	12
34	org.axonframework.common.ReflectionUtils	ReflectionUtils	[Type, File, Java, Class, ByteCode, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation19, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut56, Mark4TypeHDBSCAN131, Mark4TopAnomalyBottleneck]	12
35	org.axonframework.common.AxonNonTransientException	AxonNonTransientException	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TypeWeaklyConnectedComponent0,	11

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
			Mark4TypeLabelPropagation15, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeCoreDecomposition4, Mark4TypeMaximumKCut49, Mark4TypeLocalClusteringCoe...	
36	org.axonframework.common.configuration.ComponentBuilder	ComponentBuilder	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation33, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut25, Mark4TypeHDBSCAN69]	11
37	org.axonframework.messaging.core.MessageHandlerInterceptor	MessageHandlerInterceptor	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation86, Mark4TypeLouvainCommunity12, Mark4...	11
38	org.axonframework.messaging.core.annotation.ParameterResolver	ParameterResolver	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation14, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition7, Mark4TypeMaximumKCut8, Mark4TypeHDBSCAN121]	11
39	org.axonframework.common.TypeReference	TypeReference	[Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation23, Mark4TypeLouvainCommunity12,...	11

Table 6 - Packages that are used by multiple artifacts

This table shows the top 30 artifacts that only use a few (compared to all existing) packages of another artifact. The whole table can be found in the CSV report [ArtifactPackageUsage](#).

	artifactName	dependentArtifactName	dependentPackages	dependentArtifactPackages	packageUsagePercentage	dependentFullQualifiedPackageNames	dependentPackageNames
0	axon-tracing-opentelemetry-5.0.1	axon-messaging-5.0.1	2	57	0.035088	[org.axonframework.messaging.tracing, org.axonframework.messaging.core]	[tracing, core]
1	axon-tracing-opentelemetry-5.0.1	axon-common-5.0.1	1	13	0.076923	[org.axonframework.common]	[common]
2	axon-metrics-micrometer-5.0.1	axon-messaging-5.0.1	6	57	0.105263	[org.axonframework.messaging.core, org.axonframework.messaging.commandhandling, org.axonframework.messaging.eventhandling, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.monitoring, org.axonframework.messaging.queryhandling]	[core, commandhandling, eventhandling, processing, monitoring, queryhandling]
3	axon-test-5.0.1	axon-messaging-5.0.1	8	57	0.140351	[org.axonframework.messaging.commandhandling, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.eventhandling.processing.streaming.token, org.axonframework...	[commandhandling, core, eventhandling, eventstreaming, unitofwork, token, annotation, monitoring]
4	axon-server-connector-5.0.1	axon-eventsourcing-5.0.1	1	7	0.142857	[org.axonframework.eventsourcing.eventstore]	[eventstore]
5	axon-test-5.0.1	axon-eventsourcing-5.0.1	1	7	0.142857	[org.axonframework.eventsourcing.eventstore]	[eventstore]
6	axon-server-connector-5.0.1	axon-modelling-5.0.1	1	7	0.142857	[org.axonframework.modelling]	[modelling]
7	axon-metrics-micrometer-5.0.1	axon-common-5.0.1	2	13	0.153846	[org.axonframework.common.configuration, org.axonframework.common]	[configuration, common]
8	axon-spring-boot-autoconfigure-5.0.1	axon-server-connector-5.0.1	1	5	0.200000	[org.axonframework.axonserver.connector]	[connector]
9	axon-spring-boot-autoconfigure-5.0.1	axon-test-5.0.1	1	5	0.200000	[org.axonframework.test.server]	[server]
10	axon-modelling-5.0.1	axon-messaging-5.0.1	13	57	0.228070	[org.axonframework.messaging.commandhandling.annotation, org.axonframework.messaging.eventhandling.conversion, org.axonframework.messaging.core.configuration.reflection, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core, org.axonframework.messaging.core.unitofwork, org....]	[annotation, conversion, reflection, eventhandling, core, unitofwork, configuration, commandhandling]
11	axon-conversion-5.0.1	axon-common-5.0.1	3	13	0.230769	[org.axonframework.common.infra, org.axonframework.common, org.axonframework.common.annotation]	[infra, common, annotation]
12	axon-update-5.0.1	axon-common-5.0.1	3	13	0.230769	[org.axonframework.common.annotation, org.axonframework.common, org.axonframework.common.configuration]	[annotation, common, configuration]
13	axon-modelling-5.0.1	axon-conversion-5.0.1	1	4	0.250000	[org.axonframework.conversion]	[conversion]
14	axon-conversion-5.0.1	axon-conversion-5.0.1	1	4	0.250000	[org.axonframework.conversion]	[conversion]
15	axon-eventsourcing-5.0.1	axon-conversion-5.0.1	1	4	0.250000	[org.axonframework.conversion]	[conversion]
16	axon-spring-boot-autoconfigure-5.0.1	axon-messaging-5.0.1	15	57	0.263158	[org.axonframework.messaging.core.correlation, org.axonframework.messaging.queryhandling.distributed, org.axonframework.messaging.eventhandling.processing.streaming.token.store, org.axonframework.messaging.commandhandling.distributed, org.axonframework.messaging.core.interception, org.axonframew...	[correlation, distributed, store, interception, queryhandling, conversion, timeout, eventhandling, commandhandling, transaction, annotation, jpa, core]
17	axon-eventsourcing-5.0.1	axon-messaging-5.0.1	16	57	0.280702	[org.axonframework.messaging.eventhandling.processing.streaming.token, org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core, org.axonframework.messaging.eventstreaming, org.axonframework.messaging.core.annotation, org.axonframew...	[token, unitofwork, eventhandling, core, eventstreaming, annotation, conversion, transaction, configuration, commandhandling, interception]
18	axon-server-connector-5.0.1	axon-messaging-5.0.1	16	57	0.280702	[org.axonframework.messaging.eventstreaming, org.axonframework.messaging.eventhandling.processing.streaming.segmenting, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.core, org.axonfram...	[eventstreaming, segmenting, store, processing, core, subscribing, eventhandling, streaming, unitofwork, conversion, token, queryhandling, commandhandling, distributed]
19	axon-spring-boot-autoconfigure-5.0.1	axon-eventsourcing-5.0.1	2	7	0.285714	[org.axonframework.eventsourcing.eventstore.jpa, org.axonframework.eventsourcing.eventstore]	[jpa, eventstore]
20	axon-spring-boot-autoconfigure-5.0.1	axon-common-5.0.1	4	13	0.307692	[org.axonframework.common.configuration, org.axonframework.common.jdbc, org.axonframework.common.jpa, org.axonframework.common]	[configuration, jdbc, jpa, common]
21	axon-test-5.0.1	axon-common-5.0.1	4	13	0.307692	[org.axonframework.common, org.axonframework.common.configuration, org.axonframework.common.infra, org.axonframework.common.annotation]	[common, configuration, infra, annotation]
22	axon-modelling-5.0.1	axon-common-5.0.1	5	13	0.384615	[org.axonframework.common, org.axonframework.common.configuration, org.axonframework.common.property, org.axonframework.common.infra, org.axonframework.common.annotation]	[common, configuration, property, infra, annotation]
23	axon-common-5.0.1	axon-common-5.0.1	6	13	0.461538	[org.axonframework.common, org.axonframework.common.annotation, org.axonframework.common.lifecycle, org.axonframework.common.infra, org.axonframework.common.io, org.axonframework.common.configuration]	[common, annotation, lifecycle, infra, io, configuration]
24	axon-server-connector-5.0.1	axon-common-5.0.1	6	13	0.461538	[org.axonframework.common, org.axonframework.common.util, org.axonframework.common.infra, org.axonframework.common.configuration, org.axonframework.common.annotation, org.axonframework.common.lifecycle]	[common, util, infra, configuration, annotation, lifecycle]
25	axon-messaging-5.0.1	axon-conversion-5.0.1	2	4	0.500000	[org.axonframework.conversion, org.axonframework.conversion.json]	[conversion, json]
26	axon-metrics-micrometer-5.0.1	axon-metrics-micrometer-5.0.1	1	2	0.500000	[org.axonframework.extension.metrics.micrometer.reservoir]	[reservoir]
27	axon-eventsourcing-5.0.1	axon-common-5.0.1	7	13	0.538462	[org.axonframework.common.infra, org.axonframework.common.annotation, org.axonframework.common, org.axonframework.common.configuration, org.axonframework.common.jpa, org.axonframework.common.jdbc, org.axonframework.common.io]	[infra, annotation, common, configuration, jpa, jdbc, io]
28	axon-spring-boot-autoconfigure-5.0.1	axon-update-5.0.1	3	5	0.600000	[org.axonframework.update.detection, org.axonframework.update.configuration, org.axonframework.update]	[detection, configuration, update]
29	axon-test-5.0.1	axon-test-5.0.1	3	5	0.600000	[org.axonframework.test, org.axonframework.test.util, org.axonframework.test.matchers]	[test, util, matchers]

Table 7 - Types that are used by multiple artifacts

This table shows the top 30 types that only use a few (compared to all existing) types of another artifact. The whole table can be found in the CSV report `ClassesPerPackageUsageAcrossArtifacts` .

	artifactName	dependentArtifactName	packageName	dependentPackage.fqn	dependentTypes	dependentPackageTypes	typeUsagePercentage	
0	axon-eventsourcing-5.0.1	axon-messaging-5.0.1	org.axonframework.eventsourcing.configuration	org.axonframework.messaging.core	1	80	0.012500	[
1	axon-modelling-5.0.1	axon-messaging-5.0.1	org.axonframework.modelling.repository	org.axonframework.messaging.core	1	80	0.012500	[
2	axon-test-5.0.1	axon-messaging-5.0.1	org.axonframework.test.matchers	org.axonframework.messaging.core	1	80	0.012500	
3	axon-eventsourcing-5.0.1	axon-messaging-5.0.1	org.axonframework.eventsourcing.configuration	org.axonframework.messaging.core.annotation	1	50	0.020000	[org.axonframe
4	axon-spring-boot-autoconfigure-5.0.1	axon-messaging-5.0.1	org.axonframework.extension.springboot.autoconfig	org.axonframework.messaging.core.annotation	1	50	0.020000	[org.axonframe
5	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.core	org.axonframework.common.configuration	1	46	0.021739	
6	axon-eventsourcing-5.0.1	axon-common-5.0.1	org.axonframework.eventsourcing.annotation	org.axonframework.common.configuration	1	46	0.021739	
7	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.core.uniframework	org.axonframework.common.configuration	1	46	0.021739	[org.axonframe
8	axon-modelling-5.0.1	axon-common-5.0.1	org.axonframework.modelling.entity.annotation	org.axonframework.common.configuration	1	46	0.021739	
9	axon-eventsourcing-5.0.1	axon-common-5.0.1	org.axonframework.eventsourcing.annotation.reflection	org.axonframework.common.configuration	1	46	0.021739	
10	axon-metrics-micrometer-5.0.1	axon-messaging-5.0.1	org.axonframework.extension.metrics.micrometer	org.axonframework.messaging.core	2	80	0.025000	
11	axon-tracing-opentelemetry-5.0.1	axon-messaging-5.0.1	org.axonframework.extension.tracing.opentelemetry	org.axonframework.messaging.core	2	80	0.025000	
12	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.queryhandling.annotation	org.axonframework.common	1	34	0.029412	
13	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.core.interception.annotation	org.axonframework.common	1	34	0.029412	
14	axon-eventsourcing-5.0.1	axon-common-5.0.1	org.axonframework.eventsourcing.annotation	org.axonframework.common	1	34	0.029412	
15	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.eventstreaming	org.axonframework.common	1	34	0.029412	
16	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.eventhandling.tracing	org.axonframework.common	1	34	0.029412	
17	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.eventhandling.processing	org.axonframework.common	1	34	0.029412	
18	axon-test-5.0.1	axon-common-5.0.1	org.axonframework.test.server	org.axonframework.common	1	34	0.029412	
19	axon-metrics-micrometer-5.0.1	axon-common-5.0.1	org.axonframework.extension.metrics.micrometer	org.axonframework.common	1	34	0.029412	
20	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.queryhandling.tracing	org.axonframework.common	1	34	0.029412	
21	axon-test-5.0.1	axon-common-5.0.1	org.axonframework.test.fixture	org.axonframework.common	1	34	0.029412	
22	axon-test-5.0.1	axon-common-5.0.1	org.axonframework.test.util	org.axonframework.common	1	34	0.029412	
23	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.core.timeout	org.axonframework.common	1	34	0.029412	
24	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.eventhandling.sequencing	org.axonframework.common	1	34	0.029412	
25	axon-spring-boot-autoconfigure-5.0.1	axon-common-5.0.1	org.axonframework.extension.springboot.autoconfig	org.axonframework.common	1	34	0.029412	
26	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.commandhandling.tracing	org.axonframework.common	1	34	0.029412	
27	axon-conversion-5.0.1	axon-common-5.0.1	org.axonframework.conversion.avro	org.axonframework.common	1	34	0.029412	
28	axon-tracing-opentelemetry-5.0.1	axon-common-5.0.1	org.axonframework.extension.tracing.opentelemetry	org.axonframework.common	1	34	0.029412	
29	axon-messaging-5.0.1	axon-common-5.0.1	org.axonframework.messaging.core.uniframework.annotation	org.axonframework.common	1	34	0.029412	

Table 8 - Duplicate package names across artifacts

This table shows the top 30 duplicate package names across artifacts. They are ordered by the number of duplicates descending.

This might lead to confusion, makes importing more error prone and might even lead to duplicate classes where only one of them will be loaded by the class loader. If a package is named the same way in two or more artifacts this even allows another artifact to access package protected classes, methods or members which might not be intended.

The whole table can be found in the CSV report `DuplicatePackageNamesAcrossArtifacts` .

packageName	duplicates	artifactNames
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Table 9 - Annotated elements

This table shows 30 most used Java Annotations including some examples where they are used.

	annotationName	languageElement	numberOfAnnotatedElements	
0	jakarta.annotation.NonNull	Parameter	3382	[org.axonframework.extension.tracing.opentele org.axonframework.extension.tracing.opentele org.axonframework.extension.tracing.opentelemetry.MetadataContextGetter.get(1), org.axonframework.exte
1	jakarta.annotation.NonNull	Method	833	[org.axonframework.common.util.ExecutorServiceFactory.createExecutorService, org.axonframework.comr org.axonframework.common.ReflectionUtils.fieldsOf, org.axonframework.commo org.axonframework.commo
2	jakarta.annotation.Nullable	Parameter	399	[org.axonframework.common.StringUtils.capitalize(0), org.axonframework.common org.axonframework.common.ReflectionUtils.explicitlyUnequal(1), org.axonframework.common.lifecycle.LifecycleH org.i
3	jakarta.annotation.Nullable	Method	99	[org.axonframework.common.ReflectionUtils.declaringClass, org.axonframework org.axonframework.common.configuration.AbstractComponent.name, org.axonframework.common.conf org.axonframework.common.conf
4	org.axonframework.common.annotation.Internal	Class	75	[org.axonframework.common.configuration org.axonframework.common.configuration.InstantiatedComponentDefinition, org.axonframework org.axonframework.axonserver.connector.MetadataConverter, org.axc
5	jakarta.annotation.NonNull	Field	70	[org.axonframework.com <i>HandlerRegistration, handler, org.axonframework.common.configu</i> org.axonframework.axonserver.connector.command.AxonServerCommandBusCo org.a)
6	java.lang.FunctionalInterface	Interface	55	[org.a) <i>SqlResultConverter, org.axonframework.common.jdbc.ConnectionProvider, org.axon frame</i> SqlFunction, org.axonframework.common.util.ExecutorServiceFactory, org.axonfr org
7	java.lang.annotation.Retention	Annotation	42	[org.axonframework.common.annotation.Internal, org.axonframework.common.Priority, org.axonframework org.axonframework.modelling.annotation.InjectEntity, org.axonframework.mode org.axonframe
8	java.lang.annotation.Target	Annotation	42	[org.axonframework.common.annotation.Internal, org.axonframework.common.Priority, org.axonframework org.axonframework.modelling.annotation.InjectEntity, org.axonframework.mode org.axonframe
9	org.springframework.context.annotation.Bean	Method	36	[org.axonframework.extension.springboot.autoconfigure.AxonTimeoutAutoConfiguration.messag org.axonframework.extension.springboot.autoconfigure.AxonTimeoutAutoConfiguration org.axonframework.extension.
10	org.axonframework.common.annotation.Internal	Constructor	19	[org.axonframework.eventsourcing.event org.axonframework.messaging.eventhandling.processing.streaming.pooled.PooledStreaming org.axonframework.messaging.eventhandling.processing.streaming.pole
11	org.springframework.boot.autoconfigure.condition.ConditionalOnMissingBean	Method	18	[org.axonframework.extension.springboot.autoconfigure.JpaAutoCo org.axonframework.extension.springboot.autocor org.axonframework.extension.springboot.autoconfigure.JpaAutoConfiguration.persistenceExci
12	org.springframework.boot.autoconfigure.AutoConfiguration	Class	18	[org.axonframework.extension.springboot.autocci org.axonframework.extension.springboot.autoconfigure.Correl org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoConfiguration, org.axi
13	java.lang.annotation.Documented	Annotation	15	[org.axonframework.common.annotation.Internal, org.axonframework.mode org.axonframework.eventsourcing.annotation.EventSourcingHandler, org.axonframework.extension.springt org.axonframeworkworl
14	org.axonframework.common.annotation.Internal	Interface	14	[org.axonframework.common.jdbc.ConnectionProvider, org.axonframework org.axonframework.modelling.entity.annotation.AnnotatedEntityMetamodelFactory, org.axonframework.eventsx org.axonframework.ev
15	com.fasterxml.jackson.annotation.JsonProperty	Parameter	12	[org.axonframework.messaging.eventhandling.processing.streaming.token org.axonframework.messaging.eventhandling.processing.streaming.token org.axonframework.messaging.eventhandling.processing.strei
16	org.axonframework.common.Priority	Class	12	[org.axonframework.test.Fixtur org.axonframework.messaging.eventhandling.annotation.TimestampParameterResolverFactory, org.axonframeworkworl
17	org.axonframework.common.annotation.Internal	Method	10	[org.axonframework.messaging.eventhandling.conversion org.axonframework.messaging.eventhandling.conversion.pooled.StreamingToken org.axonframework.messaging.eventhandling.processing.streaming.token
18	org.springframework.boot.context.properties.ConfigurationProperties	Class	10	[org.axonframework.axonserver.connector.AxonServerConfiguration, org.axonframework.extension.springboot org.axonframework.extension.springboot.TokenStoreProperties, org.axonframework.extension.spr
19	org.springframework.boot.context.properties.EnableConfigurationProperties	Class	9	[org.axonframework.extension.springboot.autocci org.axonframework.extension.springboot.springb org.axonframework.extension.springboot.autoconfigure org.axonframework.extension.
20	org.springframework.boot.autoconfigure.condition.ConditionalOnProperty	Class	8	[org.axonframework.extension.springboot.autocci org.axonframework.extension.springboot.autoconfigure.AvroCon <i>figured</i> org.axonframework.extension.springboot.autoconfigure.Avro
21	jakarta.persistence.Basic	Field	8	[org.axonframework.eventsourcing.even org.axonframework.eventsourcing.eventst org.axonframework.eventsourcing.eventstore org.axonframework.eventsourcing.evi
22	org.springframework.boot.context.properties.NestedConfigurationProperty	Field	8	[org.axonframework.ext <i>MessageHandlerTimeoutProperties.commands, org.axonframework.extension.:</i> MessageHandlerTimeoutProperties.queries, org.axonframework.extension.springboot.TimeoutProperties\$MessageH
23	java.lang.SafeVarargs	Constructor	7	[org.axonframework.test.matchers.SequenceMatcher.<init>, org.axonframe org.axonframework.test.matchers.ExactSequenceMatcher.<init>, org.axonframework.test.r org.axonframework.test.matchers
24	org.springframework.boot.autoconfigure.condition.ConditionalOnClass	Class	7	[org.axonframework.extension.springboot.autoconfigure org.axonframework.extension.springboot org.axonframework.extension.springboot.autoconfigure.AxonServerAutoConfiguration, org.axonframework.extension
25	org.springframework.context.annotation.Conditional	Method	6	[org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoCo org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoConfiguratio org.axonframework.extension.springboot.au
26	org.springframework.boot.context.properties.bind.DefaultValue	Parameter	6	[org.axonframework.extension.springboot.JpaEventStorageEng org.axonframework.extension.springboot.JpaEventStorageEng org.axonframework.extension.springboot.JpaEventStorageEngineConfigura
27	com.fasterxml.jackson.annotation.JsonCreator	Constructor	6	[org.axonframework.messaging.eventhandling.processing.streaming.ti org.axonframework.messaging.eventhandling.processing.strai org.axonframework.messaging.eventhandling.processing.streaming.token
28	org.springframework.boot.autoconfigure.AutoConfigureBefore	Class	6	[org.axonframework.extension.springboot.autoconfigure.Correl org.axonframework.extension.springboot.autoconfigure org.axonframework.extension.springboot.autoconfigure.ConverterAutoConfiguration, org.axonfr
29	java.beans.ConstructorProperties	Constructor	6	[org.axonframework.messaging.eventhandling.processing.streaming.ti org.axonframework.messaging.eventhandling.processing.strai org.axonframework.messaging.eventhandling.processing.streaming.token

Table 10 - Distance distribution between dependent files

This table shows the file directory distance distribution between dependent files. Intuitively, the distance is given by the fewest number of change directory commands needed to navigate between a file and a dependency it uses. Those are aggregate to see how many dependent files are in the same directory, how many are just one change directory command apart, and so on.

dependency.fileDistanceAsFewestChangeDirectoryCommands	numberOfDependencies	numberOfDependencyUsers	numberOfDependencyProviders	
0	0	2080	854	896
1	1	96	82	41
2	2	2067	606	417
3	4	2019	650	309